



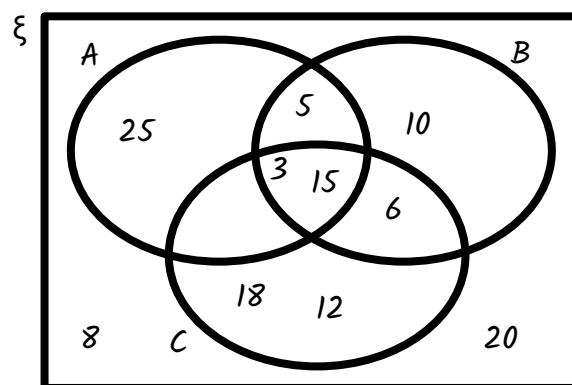
Using a Venn diagram to display outcomes for more than two events

Task A: Venn diagrams

1) A spinner has the possible outcomes of {3, 5, 6, 8, 10, 12, 15, 18, 20, 25}.

Event A = {odd numbers}, event B = {factors of 30} and event C = {multiples of 3}.

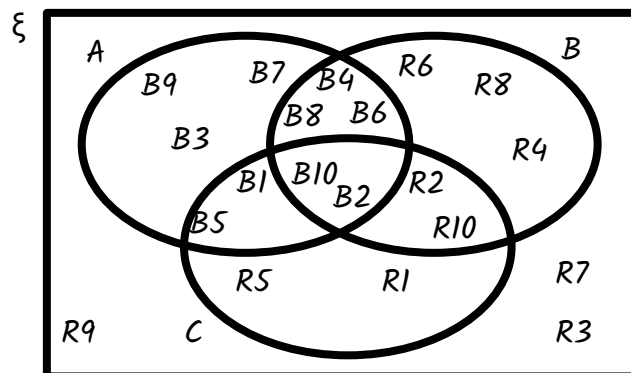
Draw a **Venn diagram** to show this information.



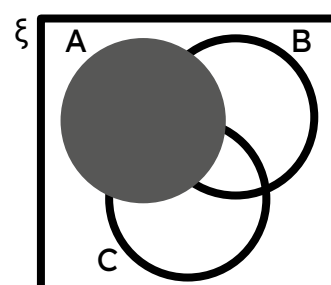
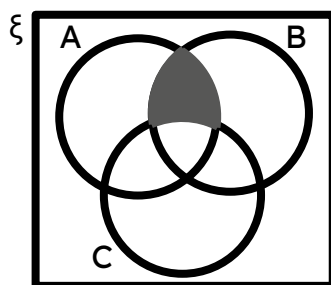
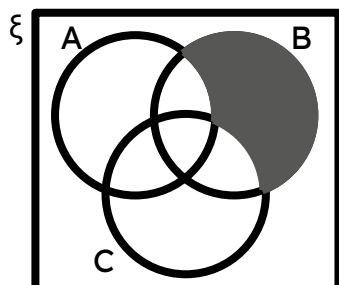
2) There are 20 playing cards. Each card is either red(R) or black(B) and a number from 1 to 10. For example, R3 is a red number 3.

Event A = {black cards}, event B = {even number} and event C = {factor of 10}.

Draw a **Venn diagram** to show this information.



3) Match the statements to the shaded areas.



outcomes of event A

outcomes of event B only

outcomes of event A and event B but not event C



Task B: Outcomes from Venn diagrams

1) Use this Venn diagram to answer the following questions:

a) List the outcomes in event A

$\{2, 4, 6, 8, 10, 12\}$

b) List the outcomes in event C

$\{1, 2, 3, 4, 6, 12\}$

c) List the outcomes in both event B and event C

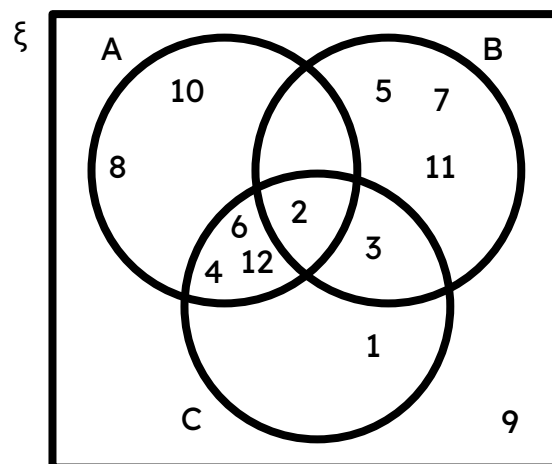
$\{2, 3\}$

d) List the outcomes that are only in event B

$\{5, 7, 11\}$

e) List the outcome that is not in any of the events

$\{9\}$



2) The Venn diagram, shows which pupils have chosen history (H), maths (M) or psychology (P) as their GCSE choices.

a) Which pupils have chosen to study history?

$\{Alex, Aisha, Jun, Sam, Sofia\}$

b) Which pupils have chosen to study history and psychology?

$\{Sofia, Sam\}$

c) Complete this section of the outcome tree

